

REAL-TIME HARMFUL OBJECT DETECTION IN SURVEILLANCE ENVIRONMENTS USING YOLOv11 DEEP LEARNING ARCHITECTURE

Omkar Rupnawar*1, Ajinkya Kadam*2, Ashish Jawane*3, Sandesh Jagdale*4

*1,2,3,4Student, Department of Computer Engineering, Pune, Maharashtra, India.

Abstract

Public safety in urban spaces has become a growing concern worldwide, and the need for intelligent surveillance mechanisms has never been more pressing. Traditional closed-circuit television setups rely heavily on human operators who are susceptible to fatigue, distraction, and delayed response times. This paper presents the design and implementation of a real-time harmful object detection system built upon the YOLOv11 (You Only Look Once, version 11) deep learning framework. The proposed system is specifically trained to identify threatening objects such as knives and bladed weapons from live camera feeds with high accuracy and minimal latency. A custom dataset comprising over three thousand annotated images of knives, scissors, and bladed instruments was curated and used for transfer learning on the YOLOv11 nano architecture. The model achieves a mean average precision (mAP@0.5) of 89.2 percent on the validation set while maintaining inference speeds of approximately 28 frames per second on consumer-grade hardware. A confidence thresholding mechanism is employed to filter out low-confidence predictions and reduce spurious detections. The complete pipeline is integrated into a desktop application featuring a graphical user interface, an SQLite event database, and an automated email alerting mechanism. Experimental evaluations demonstrate that the system attains a precision of 91.4 percent, a recall of 87.6 percent, and an F1-score of 89.5 percent, making it a viable candidate for deployment in schools, transit stations, and other public venues.

Keywords: Object Detection, YOLOv11, Deep Learning, Surveillance, Harmful Object, CNN, Real-Time Detection

I. INTRODUCTION

Ensuring public safety in crowded urban environments remains one of the foremost challenges faced by law enforcement agencies and civic administrators across the globe. According to data published by the National Crime Records Bureau, incidents involving sharp-edged weapons have shown a steady uptick over the past decade, underscoring the urgency for proactive detection mechanisms that can operate around the clock without human fatigue.

Conventional surveillance infrastructure, which predominantly consists of closed-circuit television cameras monitored by security personnel, suffers from well-documented limitations. A study conducted by Keval and Sasse in 2010 established that the attentiveness of a human operator drops significantly after roughly twenty minutes of continuous monitoring, leading to missed events and delayed reactions. The sheer volume of camera feeds in large installations such as railway stations and shopping complexes further compounds this problem, as a single operator is often expected to track dozens of screens simultaneously.

The advent of deep learning, and specifically convolutional neural network architectures designed for object detection, has opened up avenues for automating this task. Among the various detector families, the YOLO (You Only Look Once) series has gained widespread adoption on account of its favourable balance between detection accuracy and inference speed. Unlike two-stage detectors such as Faster R-CNN, YOLO performs detection in a single forward pass through the network, making it particularly well-suited for real-time applications where latency is a critical constraint.

This paper puts forward a complete end-to-end pipeline for detecting harmful objects, with a particular focus on knives and bladed instruments, from live webcam feeds. The contributions of this work are threefold. First, a custom annotated dataset of harmful objects is prepared and used to fine-tune the YOLOv11 nano model via transfer learning. Second, a confidence-based filtering mechanism is applied at inference time to discard low-scoring predictions and minimise false positives. Third, the trained model is embedded within a modular desktop application that provides a graphical interface, persistent event logging in an SQLite database, and automated email alerts upon confirmed threat detection.

II. LITERATURE REVIEW

Object detection has witnessed remarkable progress since the introduction of region-based convolutional neural networks by Girshick and colleagues in 2014. The original R-CNN pipeline extracted region proposals using selective search and classified each proposal independently, resulting in prohibitively slow inference times.

Subsequent refinements, including Fast R-CNN and Faster R-CNN, addressed these bottlenecks by sharing convolutional features across proposals and replacing selective search with a trainable Region Proposal Network, respectively. While Faster R-CNN remains a benchmark for high-accuracy detection, its two-stage architecture inherently limits throughput.

The YOLO family, introduced by Redmon and colleagues in 2016, recast object detection as a single regression problem. By dividing the input image into a grid and predicting bounding boxes and class probabilities directly, YOLOv1 achieved near-real-time detection at the cost of some localisation accuracy. Subsequent versions, YOLOv2 through YOLOv5, introduced batch normalisation, anchor boxes, feature pyramid networks, and mosaic data augmentation, each iteration narrowing the accuracy gap with two-stage detectors. YOLOv11, the latest release by Ultralytics in late 2024, further streamlines the architecture with an anchor-free detection head, an improved C3k2 backbone module, and a decoupled classification-regression design, yielding state-of-the-art results on the COCO benchmark.

Several researchers have explored the application of YOLO-based detectors to weapon and threat detection. Narejo and colleagues (2021) trained YOLOv5 on a firearms dataset and reported a mAP of 82 percent, though the study did not address bladed weapons. Bhatt and others (2021) employed YOLOv4 for knife detection in still images, achieving an accuracy of 84 percent but without real-time video integration. Gelana and Yadav (2019) combined traditional histogram-of-oriented-gradient features with a support vector machine classifier for weapon detection in CCTV footage, reporting a detection rate of 79 percent. While these studies demonstrate the feasibility of automated weapon detection, none of them utilise the latest generation YOLO architecture for bladed-weapon detection in live video feeds.

The present work builds upon these foundations by leveraging the latest YOLOv11 architecture, applying confidence-based filtering for false-positive reduction, and packaging the entire pipeline into a deployable desktop application with alerting capabilities.

III. PROPOSED METHODOLOGY

The proposed system follows a layered architecture comprising three principal stages: data acquisition and preprocessing, model training via transfer learning, and real-time inference integrated with an alert generation pipeline. Each stage is described below.

A. Dataset Preparation

A custom dataset was assembled by aggregating images from publicly available repositories including Roboflow and the Open Images Dataset, supplemented by manually captured photographs of knives, scissors, box cutters, and other bladed instruments under varied lighting conditions and backgrounds. The final dataset comprises 3,240 images, partitioned into a training set of 2,592 images (80 percent) and a validation set of 648 images (20 percent). Each image was annotated using bounding boxes in the YOLO-format text files, with a single class label designated as knife. Data augmentation techniques including horizontal flipping, random rotation of up to fifteen degrees, brightness jittering, and mosaic augmentation were applied during training to improve generalisation.

B. Model Architecture and Transfer Learning

YOLOv11 offers five model scales ranging from nano (n) to extra-large (x). Given the real-time constraints of the target deployment environment, the nano variant (YOLOv11n) was selected as the backbone. This variant contains approximately 2.6 million parameters and occupies roughly 5.4 megabytes of storage, making it suitable for consumer-grade hardware. Compared to its predecessors, YOLOv11 introduces the C3k2 module which replaces the older C2f block, enabling more efficient gradient flow and richer feature extraction with fewer computational operations. Transfer learning was employed by initialising the model with weights pretrained on the COCO dataset and fine-tuning all layers on the custom knife dataset for 120 epochs using the SGD optimiser with a learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. The input image resolution was fixed at 640 by 640 pixels. Training was conducted on a workstation equipped with an NVIDIA GeForce RTX 3060 GPU (12 GB VRAM) and required approximately four hours to complete.

C. System Integration and Alert Pipeline

The detection pipeline is embedded within a desktop application built using the Tkinter graphical toolkit in Python. A multi-threaded architecture is employed wherein separate threads handle camera capture, YOLO inference, and GUI rendering, ensuring that the user interface remains responsive at all times. During inference, a configurable confidence threshold (set to 0.80 in our deployment) is applied so that only high-confidence detections are forwarded to the alert subsystem, effectively filtering out low-scoring spurious predictions. Detection events are persisted in an SQLite database that records the timestamp, class label, confidence score, and file path of the associated snapshot. When a threat is confirmed above the threshold, the system saves a timestamped JPEG snapshot to disk and dispatches an email alert containing the snapshot as an attachment via SMTP over TLS. The email is sent asynchronously in a background thread to avoid blocking the inference loop. A configurable cooldown period prevents alert fatigue by enforcing a minimum interval of sixty seconds between successive emails for the same detection category.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Training Performance

The YOLOv11n model was trained for 120 epochs on the custom knife dataset. The training loss converged smoothly after approximately 80 epochs. On the validation set, the model achieved a mean average precision at an IoU threshold of 0.50 (mAP@0.5) of 89.2 percent and a mAP@0.5:0.95 of 67.8 percent. The best checkpoint, selected on the basis of validation mAP, was saved and used for all subsequent inference experiments.

B. Detection Accuracy

The trained model was evaluated on a held-out test set of 200 images that were not used during training or validation. The following metrics were obtained: Precision of 91.4 percent, Recall of 87.6 percent, F1-Score of 89.5 percent, and mAP@0.5 of 89.2 percent. These results compare favourably with prior work. Bhatt and colleagues (2021) reported an accuracy of 84 percent using YOLOv4, while Narejo and colleagues (2021) achieved a mAP of 82 percent with YOLOv5 on a firearms dataset. The improvement can be attributed to the architectural advances in YOLOv11, particularly the C3k2 backbone module, the anchor-free detection head, and the decoupled classification-regression branches, as well as the targeted data augmentation strategies employed during training.

C. Inference Speed

On the RTX 3060 GPU, the model achieved an average inference speed of 28.4 frames per second at an input resolution of 640 by 640 pixels. On CPU (Intel Core i7-12700H), the throughput dropped to approximately 8.5 frames per second. For the intended deployment scenario of single-camera monitoring, even the CPU throughput is adequate for near-real-time operation.

D. False Positive Analysis

To evaluate the robustness of the confidence thresholding strategy, a controlled experiment was conducted in which a sixty-minute video containing both genuine knife appearances and potential false-positive triggers (such as metallic reflections and letter openers) was processed. At the default confidence threshold of 0.80, the system produced only 4 false alerts across the entire duration. Lowering the threshold to 0.45 increased false alerts to 19, confirming that the chosen threshold strikes a practical balance between sensitivity and specificity for the target deployment environment.

E. Limitations

While the system performs well under controlled conditions, several limitations should be noted. First, the model is trained exclusively on knife and bladed-weapon imagery; it does not generalise to other threat categories such as firearms without additional training data. Second, performance degrades noticeably under poor lighting and heavy occlusion, as the model struggles to extract discriminative features from low-contrast regions. Third, the current implementation supports only a single camera feed; scaling to multi-camera deployments would require architectural modifications to the threading model and potentially GPU-level batching of inference requests.

V. CONCLUSION AND FUTURE WORK

This paper has presented a complete pipeline for real-time harmful object detection in surveillance environments, built upon the YOLOv11 deep learning architecture. The system achieves a precision of 91.4 percent and a recall of

87.6 percent on a custom knife dataset, while maintaining inference speeds suitable for live monitoring on consumer hardware. The confidence thresholding strategy employed in this work keeps false alarms to a minimum without sacrificing detection sensitivity, making the system practical for real-world deployment scenarios.

Looking ahead, several avenues for enhancement present themselves. Expanding the dataset to cover a broader range of threat objects, including firearms and blunt instruments, would increase the utility of the system. Integrating the detection pipeline with edge computing platforms such as the NVIDIA Jetson series would enable deployment in bandwidth-constrained environments. Additionally, incorporating attention mechanisms and transformer-based backbones into the detector could further improve accuracy on small and partially occluded objects. Finally, a longitudinal field trial in a real-world setting, conducted in collaboration with law enforcement stakeholders, would provide invaluable feedback for refining the system prior to large-scale adoption.

VI. REFERENCES

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